LARCH

PLANT MAINTENANCE

& OEE SOLUTION

End-to-End Asset & Equipment Management Platform

Reduced Downtime

Improved OEE

Industry 4.0 Ready

Powering Smart Manufacturing Excellence

PLATFORM MODULES

Comprehensive maintenance intelligence across your entire plant floor

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Asset & Equipment Mgmt

→ *Details inside*

Preventive Maintenance

→ *Details inside*

Breakdown Maintenance

→ *Details inside*

Predictive Maintenance

→ *Details inside*

Tooling, Die & Mold Mgmt

→ *Details inside*

Compliance & Audit Readiness

→ *Details inside*

01 ASSET & EQUIPMENT MANAGEMENT

Machine Master

Line, cell & station-wise equipment registry with OEM, model, serial & warranty details

Tooling & Die Management

Shot/cycle/hour-based tool life tracking with location & status visibility

Asset Hierarchy

Plant → Line → Machine → Sub-assembly drill-down structure

Tool Failure Analysis

Damage & failure root cause capture for continuous improvement

Criticality Classification

A/B/C classification for prioritized maintenance attention

Asset Categories

CNC, Press, Robot, Furnace, Utility & more — fully configurable

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KEY METRICS

360°
Asset Visibility

A/B/C
Criticality Model

100%
Digital Records

02 PREVENTIVE MAINTENANCE

Proactive scheduling, digital checklists & compliance tracking

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PM PLANNING & SCHEDULING

- Time-based, usage-based & condition-based PM
- OEM-recommended maintenance schedules
- Shift-wise & calendar-based planning
- Automatic PM work order generation
- PM compliance tracking & reporting

PM CHECKLISTS

Machine-Specific Checklists
Tailored tasks per equipment type

Digital Execution
Mobile/tablet-based checklist completion

Photo & Remark Capture
Visual evidence attached to each task

E-Signature & Timestamp
Tamper-proof digital audit trail

Technician Assignment
Role-based task allocation & tracking

03 BREAKDOWN MAINTENANCE

Rapid response, root cause analysis & chronic loss elimination

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STEP 1

Instant Logging

Shop floor & mobile breakdown capture with severity, priority & downtime reason coding. Machine interlock with MES for automatic line stoppage.

STEP 2

Escalation & Alerts

Automated notifications to technicians & supervisors based on breakdown severity. Configurable escalation matrix ensures no breakdown goes unattended.

STEP 3

Root Cause & CAPA

Structured root cause analysis with corrective & preventive action linkage. Repeat failure identification drives chronic loss elimination.

- MTTR Tracking
- MTBF Analysis
- Downtime Pareto
- Repeat Failure Flags

04 PREDICTIVE MAINTENANCE

Condition-based monitoring with AI-driven failure prediction and IoT integration

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CONDITION MONITORING

- Vibration analysis & trending
- Temperature & pressure monitoring
- Oil analysis data capture
- SPC-like trend analysis for equipment
- Threshold-based health alerts

INDUSTRY 4.0 — AI PREDICTION

IoT / Sensor Integration

Real-time data streams from machines to cloud dashboards

AI-Based Failure Prediction

Machine learning models detect anomalies before failure occurs

RUL Estimation

Remaining Useful Life calculated for each critical component

Maintenance Recommendation Engine

Auto-suggest maintenance actions based on predicted degradation

05 WORK ORDER & SPARE PARTS MANAGEMENT

Unified work order lifecycle with intelligent spare parts planning

PM	BM	CM	IM
Preventive	Breakdown	Corrective	Improvement

WORK ORDER FEATURES

- Technician assignment & skill tracking
- Labor hours & cost recording
- Work completion verification
- Complete maintenance history log

SPARE PARTS PLANNING

- Spare part master & BOM linkage
- Min/reorder level auto-alerts
- Critical spare identification
- Integration with inventory & purchase

06 DASHBOARDS & KPIs

Real-time visibility into overall equipment effectiveness and maintenance performance

OEE

Overall Equipment Effectiveness

- Availability
- Performance
- Quality

MTBF

Mean Time Between Failures

MTTR

Mean Time To Repair

PM %

PM Compliance Rate

Cost

Maintenance Cost vs Budget

Pareto

Downtime Reason Analysis

Spare

Spare Consumption Analysis

📊 Live dashboards | 📱 Mobile-ready | 🔗 MES integration | 🏭 Plant-wide visibility

07 COMPLIANCE & INTEGRATION

IATF 16949, TPM, MES, Quality & full audit readiness — built in

IATF 16949

Automotive Standard

TPM / 5S

Total Productive Maint.

ISO Calibration

Legal Compliance

Digital Audit Trail

E-Signature & Timestamp

INTEGRATION ECOSYSTEM



MES

Maintenance-production synchronization, auto line stoppage on critical breakdown



Quality

Machine-to-quality traceability, SPC deviation → maintenance request automation



Utility

Power, HVAC, DG, compressor & water treatment monitoring with energy KPIs



Inventory & Purchase

Spare part reorder triggers linked directly to ERP procurement workflows

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OPERATIONAL BENEFITS

Why leading manufacturers choose our platform

↓ Downtime

Reduced unplanned breakdowns through predictive & preventive actions

↑ Machine Life

Extended asset lifespan with data-driven maintenance routines

↓ Cost

Lower maintenance spend through smarter spare planning & scheduling

↑ OEE

Improved overall equipment effectiveness and throughput metrics

Audit Ready

Electronic approvals, version-controlled SOPs, complete digital trail

Industry 4.0

IoT, AI & sensor integration for the smart factory of the future

SMART MAINTENANCE · STRONGER OEE · FUTURE-READY MANUFACTURING